

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

MID SEMESTER EXAMINATION — SOLUTIONS & MARKING SCHEME

Subject: Database Management Systems (DBMS)

Time: 1.5 Hours

Semester / Programme: 4th Semester

Full Marks: 20

Instructions: Answer Question No. 1 (Compulsory) and any THREE from Question No. 2 to 5. Figures in brackets indicate marks.

SECTION - A (Compulsory)

1(a) What is data independence in DBMS? Differentiate between logical and physical data independence. [1]

Solution & Key Concept:

- Data independence is the capacity to change the schema at one level of a database system without having to change the schema at the next higher level.
- Physical data independence: Ability to modify the physical schema without changing the conceptual schema.
- Logical data independence: Ability to modify the conceptual schema without changing external schemas or application programs.

Answer: Physical independence relates to storage structures; logical independence relates to conceptual definitions/views.

Marking Rubric: [0.5 Mark] for defining data independence, [0.5 Mark] for correct distinction between logical and physical.

1(b) State the cardinality ratio and participation constraint for a binary relationship set 'Enroll' between student entity set and course entity set. [1]

Solution & Key Concept:

- Cardinality ratio specifies the maximum number of relationship instances an entity can participate in (e.g., Many-to-Many: a student can enroll in many courses, a course can have many students).
- Participation constraint specifies whether the existence of an entity depends on its being related to another entity via the relationship set (Total vs Partial).

Answer: Cardinality: M:N (Many-to-Many); Participation: Typically total for Student and partial or total for Course.

Marking Rubric: [0.5 Mark] for cardinality specification, [0.5 Mark] for participation constraint explanation.

1(c) Define super key and candidate key. Is every candidate key a super key? [1]

Solution & Key Concept:

- Super key: A set of one or more attributes that, taken collectively, allow us to identify uniquely a tuple in the relation.
- Candidate key: A minimal super key (no proper subset of the key is a super key).
- Yes, every candidate key is a super key, but not every super key is a candidate key.

Answer: Yes, all candidate keys are super keys due to superset properties.

Marking Rubric: [0.5 Mark] for definitions of super key and candidate key, [0.5 Mark] for the truth value confirmation.

1(d) What is the role of the σ (σ) and π (π) operators in Relational Algebra? [1]
 $\sigma_{condition}(R)$ and $\pi_{attributes}(R)$

Solution & Key Concept:

- $\sigma_{condition}(R)$ is the Selection operator used to select tuples that satisfy a given predicate (horizontal subset).
- $\pi_{attributes}(R)$ is the Projection operator used to select specific columns/attributes from a relation, eliminating duplicates (vertical subset).

Answer: σ filters rows based on conditions; π selects specific columns.

Marking Rubric: [0.5 Mark] for σ (Selection), [0.5 Mark] for π (Projection).

1(e) Given a relation schema $(R(A, B, C, D))$ with functional dependency set $(F = \{A \rightarrow B, B \rightarrow C\})$, compute $(A)^{+}$ [1]

Solution & Key Concept:

- Initialize closure $(X^{+} = \{A\})$.
- Since $(A \rightarrow B \text{ in } F)$, add (B) to closure. $(X^{+} = \{A, B\})$.
- Since $(B \rightarrow C \text{ in } F)$, add (C) to closure. $(X^{+} = \{A, B, C\})$.
- No more FDs can be applied.

Answer: $(A^{+} = \{A, B, C\})$

Marking Rubric: [1 Mark] for correct step-by-step attribute closure calculation resulting in $(\{A, B, C\})$.

SECTION - B (Answer any THREE)

Answer any THREE questions from Q.2 to Q.5. Each question carries 5 marks.

Q.2. (a) Explain the 3-schema architecture of a Database Management System. What is the primary objective of this architecture? [2.5]

Solution for (a):

- 1. Internal Level: Deals with physical storage structure of the database (low-level data structures, indexes, file organization).
- 2. Conceptual Level: Describes the structure of the whole database for a community of users, hiding physical storage details.
- 3. External/View Level: Closest to users, describes only that part of the database which a particular user group is interested in.
- Primary Objective: To achieve data independence (both logical and physical) and separate user applications from physical database representation.

Result / Conclusion: Internal, Conceptual, and External schemas designed to achieve data independence.

Marking Rubric: [1.5 Marks] for clear explanation of the 3 levels, [1.0 Mark] for stating the primary objective.

(b) Discuss the concepts of Specialization and Generalization in E-R modeling with suitable examples and appropriate diagrammatic notations. [2.5]

Solution for (b):

- Generalization: Bottom-up approach where multiple lower-level entity sets with shared attributes are combined into a single higher-level entity set (e.g., Car and Truck generalized to Vehicle).
- Specialization: Top-down approach where an entity set can develop distinct sub-groupings based on unique characteristics (e.g., Employee specialized into Secretary, Engineer, Manager).
- Notation: Represented by an ISA (is a) triangle connecting the higher-level and lower-level entity sets.

Result / Conclusion: Generalization is bottom-up (union); Specialization is top-down (subset derivation) using ISA relationship.

Marking Rubric: [1.0 Mark] for Generalization with example, [1.0 Mark] for Specialization with example, [0.5 Mark] for notation.

- Q.3. (a)** Consider the following relational schema for a University Database: Student (RollNo, Name, Dept, Semester) Course (CourseID, CourseName, Credits) Enroll (RollNo, CourseID, Marks) Write SQL queries for the following: (i) Find the names of all students enrolled in the course named 'Database Systems'. (ii) Find the total number of credits for courses taken by the student with RollNo 'KIIT101'. [3]

Solution for (a):

- (i) `SELECT S.Name FROM Student S JOIN Enroll E ON S.RollNo = E.RollNo JOIN Course C ON E.CourseID = C.CourseID WHERE C.CourseName = 'Database Systems';`
- (ii) `SELECT SUM(C.Credits) FROM Course C JOIN Enroll E ON C.CourseID = E.CourseID WHERE E.RollNo = 'KIIT101';`

Result / Conclusion: SQL queries utilizing JOIN operations, filtering, and aggregate functions.

Marking Rubric: [1.5 Marks] for query (i) with proper joins and condition, [1.5 Marks] for query (ii) with join and SUM aggregation.

- (b)** Explain entity integrity and referential integrity constraints with examples in relational databases. [2]

Solution for (b):

- Entity Integrity Constraint: States that primary key attributes of a relation cannot take NULL values, because primary key values identify individual tuples uniquely.
- Referential Integrity Constraint: States that a foreign key value must either match a referenced primary key value in the parent relation or be completely NULL.
- Example: In Enroll(RollNo, CourseID), RollNo must refer to a valid RollNo existing in the Student relation.

Result / Conclusion: Entity integrity restricts primary keys from being NULL; Referential integrity ensures valid foreign key mappings.

Marking Rubric: [1.0 Mark] for Entity integrity rule, [1.0 Mark] for Referential integrity rule with example.

Q.4. (a) State and prove the Pseudo-transitivity rule using Armstrong's Axioms on functional dependencies.

[3]

Formula: $X \rightarrow Y$ and $YW \rightarrow Z$ implies $XW \rightarrow Z$

Solution for (a):

- Axioms: 1. Reflexivity: If $(Y \subseteq X)$, then $(X \rightarrow Y)$. 2. Augmentation: If $(X \rightarrow Y)$, then $(XZ \rightarrow YZ)$. 3. Transitivity: If $(X \rightarrow Y)$ and $(Y \rightarrow Z)$, then $(X \rightarrow Z)$.

- Theorem (Pseudo-transitivity): If $(X \rightarrow Y)$ and $(YW \rightarrow Z)$, then $(XW \rightarrow Z)$.

• Proof Steps:

- 1. Given $(X \rightarrow Y)$
- 2. Apply Augmentation rule with (W) on both sides: $(XW \rightarrow YW)$
- 3. Given $(YW \rightarrow Z)$
- 4. Apply Transitivity on $(XW \rightarrow YW)$ and $(YW \rightarrow Z)$, yielding $(XW \rightarrow Z)$. Hence proved.

Result / Conclusion: Proof successfully completed using Augmentation and Transitivity axioms.

Marking Rubric: [1.0 Mark] for stating Armstrong's axioms, [2.0 Marks] for step-by-step deductive proof of pseudo-transitivity.

(b) Given two sets of functional dependencies $(F_1 = \{A \rightarrow B, B \rightarrow C, A \rightarrow C\})$ and $(F_2 = \{A \rightarrow B, B \rightarrow C\})$, check whether $(F_1 \equiv F_2)$ (are they equivalent?).

[2]

$$F_1^+ = F_2^+$$

Solution for (b):

- Two sets of FDs (F_1) and (F_2) are equivalent if $(F_1^+ = F_2^+)$, which means every dependency in (F_1) can be inferred from (F_2) (i.e., $(F_1 \subseteq F_2^+)$) and vice-versa $(F_2 \subseteq F_1^+)$.
- Check 1: Can (F_1) dependencies be derived from (F_2) ? $(A \rightarrow B \in F_2)$ (Yes), $(B \rightarrow C \in F_2)$ (Yes), $(A \rightarrow C)$ can be derived using transitivity from $(A \rightarrow B)$ and $(B \rightarrow C)$ in (F_2) . So $(F_1 \subseteq F_2^+)$.
- Check 2: Can (F_2) dependencies be derived from (F_1) ? $(A \rightarrow B \in F_1)$ (Yes), $(B \rightarrow C \in F_1)$ (Yes). So $(F_2 \subseteq F_1^+)$.
- Conclusion: Since $(F_1^+ = F_2^+)$, the two sets are equivalent.

Result / Conclusion: Yes, (F_1) and (F_2) are equivalent.

Marking Rubric: [1.0 Mark] for checking $(F_1 \subseteq F_2^+)$, [1.0 Mark] for checking $(F_2 \subseteq F_1^+)$ and concluding equivalence.

Q.5. (a) What is normalization? Explain the need for normalization and define First Normal Form (1NF) with a suitable counter-example. [2.5]

Solution for (a):

- Normalization: The process of decomposing relations with undesirable anomalies into smaller, well-structured relations.
- Need: To eliminate data redundancy, avoid update anomalies (insertion, deletion, modification anomalies), and ensure dependency preservation.
- 1NF Definition: A relation is in 1NF if the domains of all attributes of the relation are atomic (indivisible/single-valued), containing no nested relations or repeating groups.
- Counter-example: A table containing a multi-valued attribute like PhoneNumbers (e.g., {'9876543210', '9123456789'}) in a single cell violates 1NF atomicity.

Result / Conclusion: 1NF requires atomic attribute values; multi-valued attributes violate 1NF.

Marking Rubric: [1.0 Mark] for normalization purpose/need, [1.0 Mark] for 1NF definition, [0.5 Mark] for counter-example.

(b) Explain Second Normal Form (2NF) and Third Normal Form (3NF) with formal definitions. [2.5]
 $X \rightarrow A$

Solution for (b):

- Second Normal Form (2NF): A relation schema $\mathcal{R}$ is in 2NF if it is in 1NF and every non-prime attribute A in $\mathcal{R}$ is fully functionally dependent on the primary key (no partial dependency of a non-prime attribute on a proper subset of any candidate key).
- Third Normal Form (3NF): A relation schema $\mathcal{R}$ is in 3NF if it is in 2NF and, for every non-trivial functional dependency $X \rightarrow A$, either: 1. X is a super key of $\mathcal{R}$, or 2. A is a prime attribute (part of some candidate key).

Result / Conclusion: 2NF eliminates partial functional dependencies; 3NF eliminates transitive dependencies of non-prime attributes.

Marking Rubric: [1.25 Marks] for precise 2NF definition, [1.25 Marks] for precise 3NF definition.

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